A Relation-Augmented Embedded Graph Attention Network for Remote Sensing Object Detection

Shu Tian, Lihong Kang, Xiangwei Xing, Jing Tian, Chunzhuo Fan, and Ye Zhang

***Abstract*— Multiclass geospatial object detection in high spatial resolution remote sensing imagery (HSRI) is still a challenging task. The main reason is that the objects in HRSI are location-variable and semantic-confusable, which results in the difficulties in differentiating the complicated spatial patterns and deriving the implicitly semantic labels among different categories of objects. In this article, we propose a relation-augmented embedded graph attention network (EGAT), which enables the full exploitation of the underlying spatial and semantic relations among objects for improving the detection performance. Specifically, we first construct two sets of spatial and semantic graphs of objects–objects for object relations modeling. Second, a Siamese architecture-based embedding spatial and semantic graph attention network is designed for relations reasoning, which is implemented by introducing the long short-term memory (LSTM) mechanism into the EGAT, for learning the relations among different categories of intraobjects and interobjects. Driven by the spatial and semantic LSTM, the EGAT-LSTM can adaptively focus on the critical information of reason graphs for spatial–semantic correlation discrimination in the embedding non-Euclidean feature space. By this way, the EGAT-LSTM can effectively capture the global and local spatial–semantic relationships of objects–objects, and then produce relations-augmented features for improving the performance of object detection. We conduct comprehensive experiments on three public datasets for multiclass geospatial object detection. Our method achieves state-of-the-art performance, which demonstrates the superiority and effectiveness of the proposed method.**

***Index Terms*— Deep learning, graph convolution neural networks (GCNs), graph learning, object detection, remote sensing.**

I. Introduction

THE rapid development of high spatial resolution (HSR) remote sensing sensors has supplied us with abundant detailed information, thereby providing us with new opportunities for advancing the automatic interpretation of HSR remote sensing imagery (HSRI). In recent years, multiclass geospatial object detection has been attracting considerable attention in the field of HSRI analysis. As one of the most fundamental and challenging tasks in the field of HSRI, it also plays a vital role in a wide range of applications such as environmental monitoring, ecological protection, and traffic planning [1]–[4]. The aim of object detection is to localize different classes of objects and predict their categories, respectively [5]. However, the HRSIs are inevitably subjected to some kinds of imaging conditions, such as the illumination intensities, viewpoint variations, clutter background, and the noise interference [6]. Over the past decades, tremendous efforts have been made in developing multiclass geospatial object detection approaches in HRSI. According to feature types designed for detectors, the object detection approaches can be categorized into traditional approach and convolutional neural networks (CNNs)-based approach [2].

In the early period, a series of traditional hand-engineered approaches are developed for geospatial object detection in HRSI. These methods exploit a sliding-window-based manner, which apply hand-crafted features and classifiers on dense image grids to detect objects [7]–[9]. However, due to the weakly semantic representation ability of the manual features, the traditional methods cannot quite meet the requirements of the widely practical applications. In addition, the increasing of diversity and complex contextual pattern in HRSI also lead to the bottleneck of these traditional detection methods. With the spectacular development of CNNs in the realm of computer vision, many researchers have attempted to transfer the CNNs-based models pre-trained on large-scale off-the-shine labeled nature datasets, to detect geospatial objects from HRSI [10].

As is known to all, the implementation of the existing CNNs-based object detection methods always starts from the generation of candidate object proposals that likely contain objects using a region proposal network (RPN). Then, these candidate object proposals are classified and located via a designed region-wise subnetwork [11]. However, these methods still suffer from some limitations. First, these CNNs-based models highly depend on the bounding box annotations, which only consider the information inside the bounding box, and neglect the auxiliary role of the spatial relationship of objects–objects. Second, the standard CNNs only operate on regular Euclidean domain like images, they fail to execute localized convolutional filters and pooling operation on non-Euclidean domain, such as the irregular topological graphs derived from the relations of objects–objects. These factors hinder the further development and improvement of these CNNs-based detection approaches.

Manuscript received January 28, 2021; revised March 16, 2021; accepted April 7, 2021. Date of publication May 18, 2021; date of current version December 29, 2021. This work was supported by the National Natural Science Foundation of China under Grant 62001032 and Grant 61601035. *(Corresponding author: Lihong Kang.)*

Shu Tian and Lihong Kang are with the Beijing Remote Sensing Information Institute, Beijing 1000192, China (e-mail: gregrs@126.com).

Xiangwei Xing is with the Institute of Electronics, Chinese Academy of Sciences, Beijing 100190, China.

Jing Tian and Chunzhuo Fan are with the Department of Microwave, Beijing Remote Sensing Information Institute, Beijing 1000192, China.

Ye Zhang is with the Department of Information Engineering, Harbin Institute of Technology, Harbin 150001, China.

Digital Object Identifier 10.1109/TGRS.2021.3073269

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

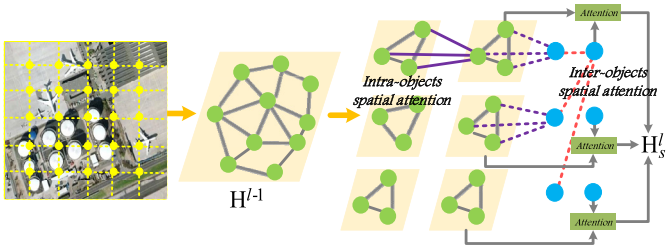


Fig. 1. Attention-based mechanism GCN.

To address these issues, several attempts have been made to introduce spatial relations into the networks by using either graphical models or spatial propagations networks [12]–[14]. In this situation, the graph convolutional networks (GCNs), as a more abstract model of nonlocal network [15], can take advantage of the spatial relations reasoning for pairwise similarity relations discrimination of objects–objects [16]. Under some circumstances, the complete relations of objects–objects are of paramount importance, particularly when some local or global object proposal regions exhibit significant appearance ambiguities. The complete relations can not only capture discriminative features in spatial configuration but also explore the semantic relationship between the set of object proposals. Meanwhile, the complete relations should also involve the contextual relations of short-range and long-range, which are helpful to infer the object–object relations under the condition of different spatial scales.

Compared with the CNNs, the GCNs is more suitable for exploiting the spatial and semantic patterns of images, and then directly mines the explicit and implicit spatial–semantic relations from the complex distributions of objects. Recent works have demonstrated that the combination of attention-based mechanism and GCNs also acts a pivotal part for capturing the global and local information, which also provides a new viewpoint for further enhancing the object detection performance [17], [18]. Moreover, multidimensional long-short term memory (LSTM) networks [19] have afforded a way to model 2-D images, where the spatial dependencies essential to object recognition can be well memorized by sequentially functioning on pixels. As shown in Fig. 1, GCNs adopt the attention mechanism that can adaptively focus on the important spatial information for deducing the implicit relations of intraobjects and interobjects. Inspired by the successes of the LSTM-based attentional models [20], [21] and the GCNs, our work aims to introduce the two-stream attentional GCNs into an embedding network, to learn the spatial and semantic relationships between interobjects and intraobjects for enhancing the performance of multiclass geospatial object detection in HRSI.

A. Motivation and Objectives

It is well-known that HRSI always contains complex and subtle spatial and semantic relations between many kinds of objects, the insufficient understanding of these contextual relationships may degrade the accuracy of object detection. The main reasons can be summarized as two aspects. On one hand, the distributions of objects in HRSI are location-variable and semantic-confusable, which also result in the difficulties in differentiating these complicated spatial layout patterns to derive the high-level semantic labels for objects detection. On the other hand, as a crucial clue, the spatial relations of objects–objects among the pairs of local and global object region proposals have rarely been attempted to explore for object detection. Therefore, the existing methods are still difficult to conquer the problems of the location diversity and the appearance ambiguity among different kinds of geospatial objects. Following these reasons, we mainly make our efforts to explore the spatial and semantic relations reasoning and discrimination of objects–objects, for enhancing the detection performance of multiclass geospatial objects in HRSI.

B. Overview and Contributions of Our Approach

In this article, we propose a novel relation-augmented embedded graph attention network (EGAT), which enables the full exploitation of the spatial and semantic attributes by introducing the LSTM-based attention mechanism for complete relationships reasoning. The LSTM is adopted to focus on the important graph-structured nodes information for enhancing the feature discriminative ability. The architecture of the proposed EGAT-LSTM framework is illustrated in Fig. 2. The whole framework consists of two spatial and semantic object relationships reasoning modules. They adopt LSTM-based mechanism for spatial and semantic relations reasoning, respectively. After that, a Siamese architecture-based EGAT is proposed for spatial and semantic similarity discrimination. As a result, the EGAT-LSTM can effectively capture the global and local spatial–semantic relationships of the interobjects and intraobjects, and then produces relations-augmented feature representations for multiclass geospatial object detection in HRSI. The contributions of our work can be summarized as follows.

1) Compared with the existing object detection works that only consider the appearance clues for features learning and discrimination, we have designed a relations-augmented EGAT for deducing the spatial–semantic relations among different kinds of intraobjects and interobjects, for enhancing the discrimination of the learned features. Moreover, due to the fact that the current GCNs and the attention-models are limited by their unexplainable attribute for some applications, we hope that the improvement of the discriminative ability by learning the embedded spatial–semantic reasoning features can also enhance the interpretability of the proposed network to some extent, which also eventually promotes the performance for multiclass object detection in HRSI.
2) An LSTM-based attention mechanism graph convolutional model is devised to model and reason the spatial–semantic relationships of objects–objects, which can effectively perform the spatial and semantic EGAT-LSTM model to adaptively focus on the important information for learning the explicit and implicit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

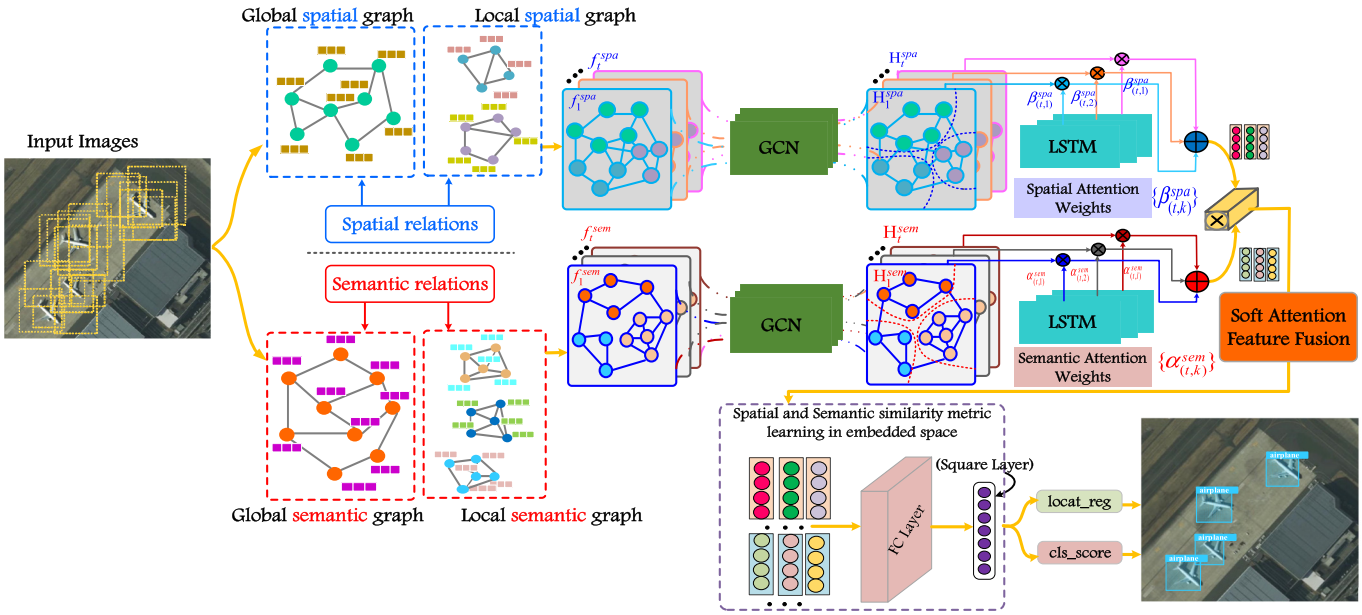


Fig. 2. Overview of our method for multiclass geospatial object detection HSRIs.

relations among different categories of interobjects and intraobjects.

3) An EGAT-LSTM structure is proposed to learn an embedded space for spatial and semantic similarity discrimination. By imposing the optimization constraints in the perspective of spatial consistency and semantic correlation, the EGAT-LSTM can produce the features with high discriminative ability, and also overcomes the problems of location diversity and appearance ambiguity for multiclass geospatial object detection.

The remainder of this article is organized as follows. Section II briefly reviews the previous work closely related to our work. Section III describes the proposed approach, including multitype domain graph construction, feature extraction, feature learning, training, and testing. The experimental results on the public datasets are reported in Section IV. Section V concludes our work and briefly discusses future work.

II. Related Work

Object detection has been investigated by a wide variety of approaches in recent years. In this section, we briefly review the existing geospatial object detection approaches, attention-based models, and graph convolutional networks.

A. Geospatial Object Detection in HRSI

According to the feature types, multiclass geospatial object detection methods can be divided into the traditional methods and the CNNs-based methods. In traditional methods, many kinds of hand-crafted features are usually designed for the task of object detection in HRSI, such as the scale invariant feature transform (SIFT) [22], the histogram of oriented gradients (HOGs) feature descriptor [23], Bag-of-words (BoW) model [24], sparse coding [25], and so on. For example, Dalal *et al.* [23] used the HOG feature and disparity maps to detect airborne vehicles in dense urban areas. Han *et al.* [26] proposed to detect multiclass geospatial objects through integrating visual saliency modeling and the discriminative learning of sparse coding. Although those hand-crafted features have shown their impressive success in some specific object detection tasks, they still have certain limitations. The main reason is that these traditional methods always lack the effective expressions of the high-level semantic information for object detection in HRSI.

In 2006, a breakthrough in deep learning was made by Hinton and Salakhutdinov [27]. Since then, the deep learning methods gradually take over the traditional hand-engineered approaches for object detection in the field of remote sensing. Compared with the traditional methods, CNNs-based approaches are capable of capturing powerful semantic feature representations by learning the deep architectures, which have been proved to effectively improve the performance of remote sensing object detection [28]. For example, Cheng *et al.* [29] proposed a rotation-invariant CNNs model by introducing and learning a new rotation-invariant layer for remote sensing object detection. Zhang *et al.* [30] constructed an iterative weakly supervised learning framework, which is combined with the candidate RPN and a localization network to detect objects in HRSIs. Zhu *et al.* [31] extracted CNNs-based features from an additional region to capture the contextual information of a candidate bounding box, and utilized an MRF inference framework to exploit object detection. Long *et al.* [32] presented an unsupervised score-based bounding box regression for accurate object localization in remote sensing images. Fu *et al.* [33] proposed a deconvolutional single shot (DSSD), which integrated a skip connection with the deconvolutional layers to improve representation capacity. Although the CNNs-based detection methods have achieved some degree of success, they are still facing some challenges for the widely practical application demands in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

field of remote sensing. The main reason is that these methods highly depend on the information inside the object proposals for learning the deep network, which neglect the auxiliary role of spatial and semantic information outside the object proposals for object detection. Therefore, they may even cause misdetections between the objects which belong to different classes but look very similar in appearance factor.

B. Attention-Based Models for Geospatial Object Detection

Recently, many studies have proved that the attention model mainly takes advantage of image-level labels to capture the global and local semantic information of object regions, which is also helpful to localize discriminant object regions for further enhancing object detection performance [34]. The key idea of attention mechanism is to assign different weights to different image regions according to their importance. This mechanism is advantageous to object detection because it adaptively focuses on the relevant regions while ignoring the irrelevant regions [35]. By this way, the attention models-based detection approaches can effectively capture the spatial and semantic relationships of intraobjects and interobjects, and then eventually boost the performance for object detection.

Some recent works have explored some new architectures to investigate the attention model-based object detection approaches, which have been achieved impressive detection performance in HRSI. For example, Wang *et al.* [36] applied the skip-connected encoder-decoder model to extract multiscale features in different layers, and made use of the multiscale feature to learn the visual attention network. This method can effectively decrease the influence of cluttered background and ensure the accuracy for multiscale geospatial object detection. Ying *et al.* [37] presented a multiattention feature pyramid network (MA-FPN) for object detection. The MA-FAN used a global spatial attention module to fuse the features obtained from the shallow layer and deep layer, to enhance the feature description ability of object position information. Then, it utilized a pixel attention module to achieve pixel-level attention maps with good details of geospatial objects, which can effectively improve the performance of object detection in HRSI. Sumbul *et al.* [38] constructed a novel multisource region attention network (MRAN) that simultaneously learns to attend regions of source images that likely to contain the interest objects. Then, the MRAN detected and recognized the fine-grained objects by leveraging an attention-driven multisource deep feature representation.

C. Graph Convolutional Networks

Graph convolutional networks (GCNs) have come up primarily to fulfill the increasing demands for irregular domain data like graphs which standard CNNs models are incapable of handling. At the present stage, the implementation of GCNs-based deep learning approaches can be roughly categorized into spectral-based and spatial-based approaches [39]. The spectral-based approaches perform graph convolutional operation on spectral domain by using the eigenvectors of the graph Laplacian as the transformation matrix. However, the operations of spectral-based approaches are sensitive to the heterogeneous data which are defined by topological graph matrix. It means that this kind of convolutional operation on graph representations can happen only if all the sample graphs share a common structure. To overcome this limitation, spatial-based graph convolutional approaches have been proposed to expand the scope of application. These methods directly perform the convolution in the graph domain by aggregating the information of neighbor nodes. By adopting appropriate strategies, the computation can be performed in a batch of nodes instead of the whole graph, which has the potential ability to improve the learning efficiency. Inspired by the success of the GCNs, some GCNs-based models have been explored in some remote sensing tasks. For example, Khan *et al.* [40] used the GCNs to extract discriminative features from the presegmented irregular local scene regions, and then, they adopted the multilabel classification approach for remote sensing scene recognition. Nevertheless, the existing GCNs-based methods merely execute the graph convolutional operation on the spatial topology graph that is constructed from the image; they ignored the exploration for semantic attributes of graphical representations, and then overlooked the spatial and semantic diversity discrimination for multiclass geospatial object detection in HRSI.

III. Proposed Framework

The proposed multiclass object detection framework starts with a graph-based object relation modeling module. We first leverage the RPN [41] to generate high-quality object region proposals, and then, the spatial and semantic relations of objects–objects can be modeled by the region adjacency graph (RAG). Constructing these RAGs are depended on the spatial consistency and semantic correlations between the pre-extracted pairs of object region proposals, which are the input for training the network model. Second, driven by the LSTM [42] attention mechanism, two Siamese structure-based GCNs are jointly embedded into the discriminative space for feature similarity discrimination and feature integration. By this way, we can learn a latent embedding space that pulls spatially and semantically coherent samples closer while pushing dissimilar samples far apart, which is helpful to enhance the ability for representing spatial–semantic relations of objects–objects. In final, the learned features with highly discriminative capability are fed into the sibling classification layer and regression layer in the fully connected layer. Thereinto, the classification layer outputs a score for each class and the regression layer predicts the corresponding precise object localization.

A. Spatial-Based Graph Convolutional Operation

GCN is a general and effective framework for learning the representation of graph-structured data. Several GCN-based variants have achieved the impressive results on many visual tasks [43], [44]. As an extension of GCN, the graph attention network (GAT) has explored more superior performance for some classification and object detection tasks [45]. Intrinsically, the GAT is a spatial-based graph convolution network,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF NOTATIONS ADOPTED IN THE PROPOSED EGAT-LSTM

Symbol	Definition
GCN notation	
$\mathcal{V}, \mathcal{E}, \mathcal{G}$	Node set, edge set and the graph
$\mathbf{A}$, $\mathbf{A}_k$	The adjacency matrix, $\mathbf{A}, \mathbf{A}_k \in \mathbb{R}^{N\times N}$
$\mathcal{B}(v_i)$	Neighbor set of the node v_i
$p(v_i, v_j)$	Minimum path length from node v_i to node v_j
$\mathbf{Z}(v_i)$	Feature vector of node v_i
ℓ	A graph labeling function for partitioning the neighbor set into a fixed number of K subsets
$Q(\cdot)$	The weight function of GCN
$C_i(v_j)$	Normalization term
f_G, f_S	The transformed spatial and semantic features of the pre-extracted object proposal regions.
(x_i, y_i, w_i, h_i)	The top-left point, the width and high of *i*-th object proposal ***r***
d_g	Dimension of the semantic features.
$\mathbf{M}_u, \mathbf{M}_v$	The projection matrices
$\mathbf{W}_G, \mathbf{W}_m, \mathbf{W}_n$	Transformed matrix for construction spatial and semantic relation graphs.
$\mathbf{A}_{spa}^{relation}, \mathbf{A}_{sem}^{relation}$	Spatial and semantic relations graphs.
Spatial and Semantic Relations based LSTM training notation	
$\mathbf{U}_s^{spa}, \mathbf{W}_{xs}^{spa}, \mathbf{W}_{hs}^{spa}$	Learnable weighted matrixes in the spatial attention module
$\mathbf{U}_s^{sem}, \mathbf{W}_{xs}^{sem}, \mathbf{W}_{hs}^{sem}$	Learnable weighted matrixes in the semantic attention module
$b_s^{spa}, b_u^{spa}, b_s^{sem}, b_u^{sem}$	Bias terms in spatial and semantic attention module, respectively
$\tilde{\mathbf{H}}_{t-1}^{spa(s)}, \tilde{\mathbf{H}}_{t-1}^{sem(s)}$	The hidden variable at *t*-1th layer of spatial and semantic LSTM model, respectively
a_t^{spa}, β_t^{sem}	Attentions scores in the spatial and semantic attention module, respectively
$\mathbf{F}_t^{spa}, \mathbf{F}_t^{sem}$	Aggregated features in output layer of the spatial and semantic LSTM network.
$\mathbf{S}_{tk}$	The spatial-semantic fused features in the output layer of EGAT-LSTM
EGAT-LSTM optimizing notation	
$\mathcal{L}_{cls}, \mathcal{L}_{reg}, \mathcal{L}_{hyb}$	The classification loss function, regression loss function, and the hybrid loss function.
$\mathbf{O}_i = (o_1, o_2, ...o_C)$	Ground-truth labels
μ, η, γ	The parameters of the regularizations, and the learning rate.
$\mathrm{smooth}_{L_1}(\cdot)$	Smooth loss function

which performs the attention mechanism to learn more discriminative features by adaptively focusing on the weights of important nodes. Herein, we first briefly review the rationale of the spatial-based graph convolutional operation. The summary of notations used in our proposed method is listed in Table I.

In the proposed method, each object relation graph is constructed on top of any region-based detector. In here, we adopt the RPN to prefetch an abundance of multiclass object region proposals, which can be considered as the set of graph nodes. We define an undirected graph $\mathcal{G} = \{(\mathcal{V}, \mathcal{E}),\ \mathbf{A}\}$, which can be encoded by the set of object proposal regions. $\mathcal{V}$ denotes the node set with $|\ \mathcal{V}\ | = N$, and $\mathcal{E}$ is the edge set of the graph, $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the adjacency matrix. On this graph, let $\mathcal{B}(v_i) = \{v_j | p(v_i, v_j) \le P\}$ be a neighbor set of the node v_i, where $p(v_i, v_j)$ is the minimum path length from v_i to v_j.

A graph labeling function $\ell : \mathcal{V}_i \to \{1, 2, \ldots, K\}$ is also designed to assign the numeric labels to each graph node, which can partition the neighbor set into a fixed number of K subsets. Therefore, the graph convolution is generally computed as

$$\mathbf{Z}_{\text{out}}(v_i) = \sum_{v_j \in \mathcal{B}} \frac{1}{C_i(v_j)} \mathbf{Z}(v_j) Q(\ell(v_j)) \tag{1}$$

where $\mathbf{Z}(v_i)$ is the feature vector of node v_i, $Q(\cdot)$ denotes the weight function that allocates the weights indicated by the labeling function $\ell(v_i)$. $C_i(v_j) = |\{v_k | \ell_i(v_k) = \ell_i(v_i)\}|$ denotes a normalization term, which aims to equal the cardinality of the corresponding subsets, and also balances their relative importance for feature learning.

Essentially, the graph convolution in the spatial domain is equivalent to average the inner product between the feature vectors of graph nodes and the corresponding weight values. Therefore, with the adjacency matrix, the spatial-based graph convolutional operation can be formulated as

$$\mathbf{Z}_{\text{out}} = \sum_{k=1}^{K} \Lambda_k^{1/2} A_k \Lambda_k^{1/2} \mathbf{Z} W_k \tag{2}$$

where $\mathbf{A}_k$ is the adjacency matrix in spatial configuration of the label $k \in \{1, 2, \ldots, K\}$, $\Lambda_k^{ij} = \sum_j A_k^{ij}$ denotes the degree matrix, and $\mathbf{W}_k$ denotes the learned weights. In here, we assume that the connections between the pairs of graph nodes are governed by $\mathbf{A}_k$, which represent the relations of objects–objects, and also can be adaptively to learn for relations reasoning. This allows the feature concatenation on the graph nodes to integrate more comprehensive relations.

B. Object Relation Graph Construction and Nodes Embedding

In order to mine the complete relationships of objects–objects, we propose two spatial and semantic relation modules, each of them is aimed to model the relations by encoding the pairs of object proposals into spatial and semantic graph representations, respectively. Therefore, we first adopt the RPN algorithm for prefetching the set of object proposal regions, which are also the input for features embedding on the nodes of the constructed graphs. The construction of relation graph is depended on how related between each pair of object proposals in their unified embedding feature space. Then, we adopt two kinds of features embedding for spatial and semantic relation graphs construction, respectively. For the spatial graphs, we utilize a bounding box regression-based feature embedding for descripting the object proposals. Each spatial-based embedded feature is corresponded to a 4-D spatial coordinate [46], which consists of the coordinate of the top-left point of the proposal, and the aspect ratio of the region proposal. For the semantic graphs, we directly choose the region of interest (ROI) features for features embedding, which can be derived from the set of convolutional operations on the visual appearance of the region proposals.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

To determine whether there is a connection between two graph nodes and how strong is the connection is, we utilize the dot product function to estimate the feature similarity of the two graph nodes, which can project the original features into subspaces to measure how strong the connection between each pair of nodes. Afterward, the corresponding adjacency relation graphs $\mathbf{A}_{\text{spa}}^{\text{relation}}$ and $\mathbf{A}_{\text{sen}}^{\text{relation}}$ can be computed based on the similarity of the embedded spatial and semantic features, respectively. The details for deriving the spatial and semantic adjacency graphs are introduced in Section III-B.

Object Spatial Relation Modeling and Node Features Embedding: To ensure that the embedded features are invariant to translation and scale transformations, we define the geometric features for each pair of object proposal region (r_i, r_j) as follows:

$$f_G(r_i, r_j) = \left\{ \log\left(\frac{|x_i - x_j|}{w_i}\right),\ \log\left(\frac{|y_i - y_j|}{h_i}\right),\ \log\left(\frac{w_j}{w_i}\right),\ \log\left(\frac{h_j}{h_i}\right) \right\}^T \tag{3}$$

where $f_G(r_i, r_j)$ denotes a 4-D geometry features of the proposal region, (x_i, y_i) is the coordinate of top-left point of the ith object proposal region, w_i and h_i denote the width and height of the ith object proposal region, respectively.

Then, the geometric feature matrix should be transformed into the nonlinear scalar matrix, and then projects it into a subspace to measure the similarity between each pair of object proposals by multiplying a geometric weight matrix $\mathbf{W}_m$ as

$$\mathbf{W}_G = \text{ReLU}\left(\mathbf{W}_m f_{G(i,j)}(r_i, r_j)\right). \tag{4}$$

After that, the spatial relation graphs of object proposals can be realized by measuring the relative geometric position as

$$\mathbf{A}_{\text{spa}(i,j)}^{\text{relation}} = \max\left\{0,\ \mathbf{W}_G \cdot f_{G(i,j)}(r_i, r_j)\right\} \tag{5}$$

where $\mathbf{A}_{\text{spa}(i,j)}^{\text{relation}}$ is taken as the adjacency matrix for spatial relation graphs of the pre-extracted object proposals, the zero trimming operation restricts the relations between objects with certain geometric structures.

Object Semantic Relation Construction and Node Features Embedding: As an important semantic cue, the semantic graph is also critical for relations reasoning of objects–objects. To derive the richer semantic relations from the pairs of object proposals, we concatenate the features extracted from the pairs of object proposal regions, and then transform them into another high-dimensional space for semantic relation graphs construction. Specifically, for the extracted semantic feature matrix which are aggregated from the ROI pooling algorithm [41], each pair of connected graph nodes go through the following transformed function as

$$f_{S(i,j)} = \text{ReLU}\left(\sum_{j \in \mathcal{N}_i} \mathbf{W}_n \left[f_{S(i)} || f_{S(j)} \right]\right) \tag{6}$$

where $f_{S(i)}$ and $f_{S(j)}$ denote the semantic features of each pairs of object proposals, and $||$ is the concatenation operation. $\mathbf{W}_n$ denotes the learnable parameter matrix, and $\mathcal{N}_i$ is the neighborhood of the ith object proposal.

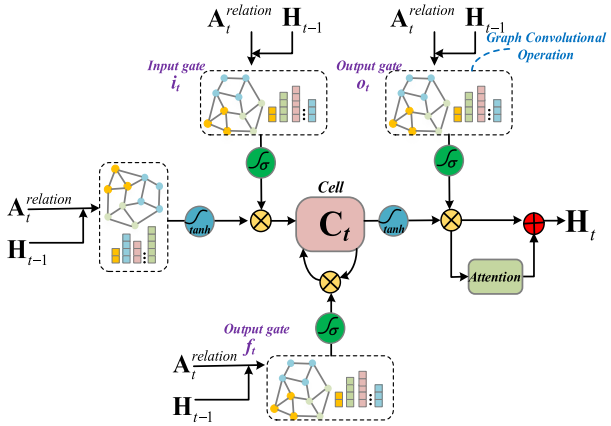


Fig. 3. Structures of EGAN-LSTM unit. Compared with LSTM, the inner operator of EGAN-LSTM is graph convolutional calculation.

Afterward, the semantic relation graph can be achieved by semantic similarity evaluation in the embedded subspace and formulated as the scaled dot product

$$\mathbf{A}_{\text{sem}(i,j)}^{\text{relation}} = \frac{\left(M_u f_{S(i)}\right)^{\text{T}} \cdot M_v f_{S(j)}}{\sqrt{d_g}} \tag{7}$$

where $\mathbf{A}_{\text{sem}(i,j)}^{\text{relation}}$ denotes the semantic relation graph of objects–objects. $\mathbf{M}_u$ and $\mathbf{M}_v$ denote the projection matrices, the role of them is to transform the weighted semantic features $f_{S(i)}$ and $f_{S(j)}$ into the subspace, and then measures the similarity between the pairs of semantic features, d_g is the dimension of the semantic features.

C. LSTM Mechanism-Driven Embedded GAN for Object Relations Reasoning

Based on the constructed spatial and semantic graphs, we proposed a Siamese-based architecture embedding GAT (EGAT) for relation reasoning. This allows each EGAT-LSTM to automatically explore and exploit the different degrees of importance of nodes in the constructed relation graphs of objects–objects, and then assign corresponding attention weights based on the LSTM-based spatial and semantic similarity discrimination. Therefore, the EGAT-LSTM can not only capture discriminative features in the spatial domain but also explore the semantic relationships between the pairs of objects for relations reasoning.

In this section, we first introduce the architecture of the proposed EGAT-LSTM whose internal operational mechanism of the unit is shown in Fig. 3. Similar to the LSTM, the unit of our EGAT-LSTM is also composed of three gates including an input gate $\mathbf{i}_t$, a forget gate $\mathbf{f}_t$, and an output gate $\mathbf{o}_t$. The difference is that these three gates are derived from the graph-structured data on the graph convolutional operator.

Based on the deduced spatial relation graph $\mathbf{A}_{\text{spa}}^{\text{relation}}$ and semantic relation graph $\mathbf{A}_{sem}^{\text{relation}}$, we propose the spatial attention module and semantic attention module respectively, which are both build on the EGAT-LSTM network. The illustrations of the spatial attention network and the semantic attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

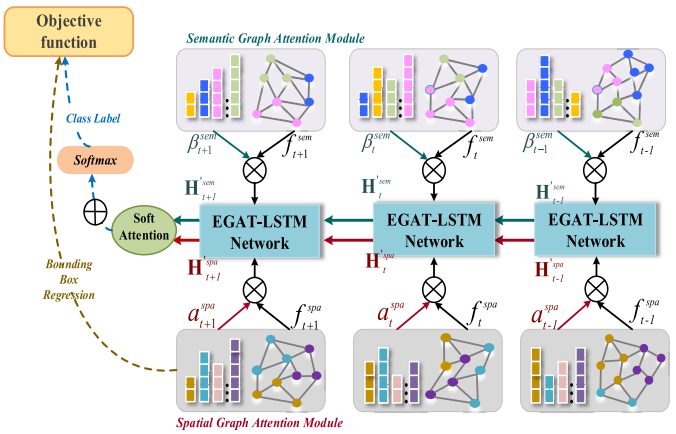


Fig. 4. Framework of the spatial and semantic structured-graph features fusion from the EGAT-LSTM network for embedded learning.

network are shown in Fig. 4. In each stream of the EGAT-LSTM, the intermediate hidden states $\tilde{\boldsymbol{H}}_t^{\text{spa}}$ and $\tilde{\boldsymbol{H}}_t^{\text{sem}}$ contain rich spatial structure information and semantic information, respectively, which are also helpful to select the important graph nodes.

Herein, the attention scores for indicating the importance of the spatial graph nodes can be obtained as

$$a_t^{\text{spa}} = \mathbf{U}_s^{\text{spa}} \tanh\left(\mathbf{W}_{xs}^{\text{spa}} \mathbf{A}_{\text{spa}(t)}^{\text{relation}} + \mathbf{W}_{hs}^{\text{spa}} \tilde{\boldsymbol{H}}_{t-1}^{\text{spa}(s)} + b_s^{\text{spa}}\right) + b_u^{\text{spa}} \tag{8}$$

where $\mathbf{U}_s^{\text{spa}}$, $\mathbf{W}_{xs}^{\text{spa}}$, and $\mathbf{W}_{hs}^{\text{spa}}$ are the learnable parameter matrices, b_s^{spa} and b_u^{spa} are the bias vectors. $\tilde{\boldsymbol{H}}_{t-1}^{\text{spa}(s)}$ denotes the hidden states at the $t-1$layer of EGAT-LSTM. As a result, the graph-selection gate of the t-th spatial graphs can be computed as a normalized scores activation function

$$a_{(t,k)}^{\text{spa}} = \frac{\exp\left(a_{(t,k)}^{\text{spa}}\right)}{\sum_{i=1}^{K} \exp\left(a_{(t,i)}^{\text{spa}}\right)}. \tag{9}$$

The set of gates control the amount of information of each spatial graph to flow to the main EGAT-LSTM network, the larger activation values denote the large attention weights for assigning the degrees of importance to the graph nodes. Then, the feature aggregation $\mathbf{F}_t^{\text{spa}}$ of the output layer of the main EGAT-LSTM network can be modulated as the weighted sum of the focused graph nodes

$$\mathbf{F}_t^{\text{spa}} = \sum_{i=1}^{K} a_t^{\text{spa}} \cdot \tilde{\boldsymbol{H}}_{ti}^{\text{spa}}. \tag{10}$$

Similar to the way of computing spatial-based attention scores, the attention scores in the semantic attention module are also can be calculated as

$$\beta_t^{\text{sem}} = U_s^{\text{sem}} \tanh\left(W_{xs}^{\text{sem}} A_{\text{sem}(t)}^{\text{relation}} + W_{hs}^{\text{sem}} \tilde{H}_{t-1}^{sem(s)} + b_s^{\text{sem}}\right) + b_u^{\text{sem}} \tag{11}$$

where $\mathbf{U}_s^{\text{sem}}$, $\mathbf{W}_{xs}^{\text{sem}}$, and $\mathbf{W}_{hs}^{\text{sem}}$ are the learnable parameter matrixes, b_s^{sem} and b_u^{sem} are the bias vectors. $\tilde{\boldsymbol{H}}_{t-1}^{\text{sem}(s)}$ denotes the hidden states at the $t-1$ layer of EGAT-LSTM. For the kth semantic graph, we regard the activation as the graph-selection function, which is calculated as

$$\beta_{(t,k)}^{\text{sem}} = \frac{\exp\left(\beta_{(t,k)}^{\text{sem}}\right)}{\sum_{i=1}^{K} \exp\left(\beta_{(t,i)}^{\text{sem}}\right)} \tag{12}$$

where the activation $\beta_{(t,k)}^{\text{sem}}$ refers to normalized attention score. Among the spatial or semantic graphs, the more important this graph is for determining the class of object.

Then, the aggregated feature $\mathbf{F}_t^{\text{sem}}$ in the semantic relation graph module also can be modulated as

$$\mathbf{F}_t^{\text{sem}} = \sum_{i=1}^{K} \beta_{ti}^{\text{sem}} \cdot \tilde{\boldsymbol{H}}_{ti}^{\text{sem}}. \tag{13}$$

In the main EGAT-LSTM network, the important node information of the constructed spatial and semantic graphs are dominated by the set of spatial and semantic gates $\tilde{\boldsymbol{H}}_{ti}^{\text{spa}}$ and $\tilde{\boldsymbol{H}}_{ti}^{\text{sem}}$, respectively. Bridging by the spatial and semantic graph-selection gate, the main EGAT-LSTM network, the spatial attention relation module, and the semantic attention relation module, are jointly trained to explicitly and implicitly learn the rich spatial and semantic information for relations reasoning of objects–objects, respectively.

D. Spatial and Semantic Features Embedding for Learning EGAT-LSTM

After the spatial and semantic relationships reasoning in the main EGAT-LSTM, we aggregate the embedded features from the LSTM mechanism-based spatial and semantic attention (SSA) modules. Hence, we design the soft attention-based spatial–semantic feature fusion function to concatenate the final features as follows:

$$\mathbf{S}_{tk} = \frac{\mathbf{F}_t^{\text{spa}} \cdot \exp\left(\mathbf{F}_t^{\text{sem}}\right)}{\sum_{k=1}^{C} \mathbf{F}_{\text{spa}(tk)}^{\text{spa}} \cdot \exp\left(\mathbf{F}_{\text{sem}(tk)}^{\text{sem}}\right)} \tag{14}$$

where the hybrid feature $\mathbf{S}_{tk}$ of objects–objects is denoted as the weighted sum of the spatial concatenated features $\mathbf{F}_t^{\text{spa}}$ and the semantic concatenated features $\mathbf{F}_t^{\text{sem}}$.

Finally, we utilize the fused hybrid features for multiclass object detection. Based on the fused features, we design a supervised hybrid-attention objective function as follows:

$$\mathcal{L}_{\text{hyb}}\left(\{O_i\}, \{F_t^{\text{spa}}\}\right) = \frac{1}{N_{\text{cls}}} \sum_i \mathcal{L}_{\text{cls}}\left(o_i, o_i^*, a_{(t,k)}^{\text{spa}}, \beta_{(t,k)}^{\text{sem}}\right) + \lambda \frac{1}{N_{\text{reg}}} \sum_i \mathcal{L}_{\text{reg}}\left(f_i^{\text{spa}}, f^{*\text{spa}}{}_i\right) \tag{15}$$

where this supervised objective function contains classification loss $\mathcal{L}_{\text{cls}}$ and regression loss $\mathcal{L}_{\text{reg}}$. The classification loss $\mathcal{L}_{\text{cls}}$ is a regularized cross-entropy function which can be defined as

$$\begin{aligned}\mathcal{L}_{\text{cls}}&\left(o_i, o_i^*, a_{(t,k)}^{\text{spa}}, \beta_{(t,k)}^{\text{sem}}\right) \\ &= -\sum_{i=1}^{C} o_i \log \tilde{o}_{ti} \\ &\quad + \mu \sum_{k=1}^{K} \left(1 - \frac{\sum_{t=1}^{T} a_{(t,k)}^{\text{spa}}}{T}\right) + \frac{\eta}{T} \sum_{t=1}^{T} \left\| \sum_{t=1}^{T} \beta_{(t,k)}^{\text{sem}} \right\|_2\end{aligned} \tag{16}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{O}_i = (o_1, o_2, \ldots, o_C)$ denotes the ground-truth label. If it belongs to the ith class, then $o_i = 1$ and $o_i = 0$ for $j \neq i$. $\tilde{y}_{ti}$ is the predicted probability of all kinds of graph-structured objects, where $\tilde{o}_{ti} = e^{\mathbf{E}ti} / \sum_{j=1}^{C} e^{\mathbf{E}ti}$. The scalars μ and η denote the two-weight decaying coefficients of the regularizations.

The first regularization term of $\mathcal{L}_{\text{cls}}$ is designed to enhance the spatial attention module to focus on more important nodes of spatial graphs. This ensures it fall into local optimum by ignoring some key graph nodes. In the second regularization term, we impose the l_2-norm on the semantic attention values to curb the unrestricted increasing of them. This can effectively alleviate gradient vanishing in the back propagation, and also reduces the overfitting for learning semantic attention module.

The formulation of the regression loss $\mathcal{L}_{\text{reg}}$ is defined as

$$\mathcal{L}_{\text{reg}}\left(f_{G(i)}, f^*_{G(i)}\right) = \sum_{i \in \{x,y,w,h\}} \text{smooth}_{L_1}\left(f_{G(i)} - f^*_{G(i)}\right). \quad (17)$$

Here, $\text{smooth}_{L_1}(\cdot)$ is the smooth L_1 loss function whose operation can be formulated as [47]

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \quad (18)$$

where $f_{G(i)}$ represents a four parameterized feature vector of the coordinates of the predicted graph-structured object proposal, the feature representation is defined in (3). $f^*_{G(i)}$ is the spatial feature vector that represents a four parameterized coordinate of each ground-truth bounding box. With the guide of labeled graph-structured graphs deduced from the pairs of object proposals, we can optimize the proposed EGAT-LSTM model via back propagation, where the embedding spatial–semantic graphs are utilized to perform gradient descent in each iteration for updating the network parameters. The implementation details of our EGAT-LSTM are shown in Algorithm 1.

Algorithm 1: EGAT-LSTM

Input: The object proposals from RPN, the maximum epoch, the learning rate γ, weight decaying coefficients μ and η.

Step 1: ***Spatial/Semantic Relation Graph Construction***
Construct the initial spatial relation $\mathbf{A}_{spa}^{relation}$ according to Eq. (1) $\sim$ Eq. (5);
Construct the initial semantic relation $\mathbf{A}_{sem}^{relation}$ according to Eq. (6) $\sim$ Eq. (7);
Step 2: ***Training the EGAT-LSTM***
while *training epoch is less then num_of_epochs* **do**
  for $t = 1$ to T **do**
    1) Calculate $\mathbf{F}_t^{spa}$ spatial attention scores α_t^{spa} and the hidden states H_t^{spa} by Eq. (8) $\sim$ Eq. (10);
    2) Calculate $\mathbf{F}_t^{sem}$ spatial attention scores β_t^{sem} and the hidden states H_t^{sem} by Eq. (11) $\sim$ Eq. (13);
  end
  Spatial-Semantic features $\mathbf{S}$ ***tk*** aggregation using Eq. (14);
end
Step3: ***Calculate the loss sfunction***
Calculate the classification loss and localization loss with attention weight using Eq. (16) $\sim$ Eq. (18);
Back-propagate
loss gradient for updating the spatial and semantic weight matrix $\mathbf{W}$ using Eq. (15);
if *convergence* **then**
  break loop;
end if
Output: Multiclass object detection results.

IV. Experiments

The section provides a concise description of the experimental details such as datasets, evaluation metrics, implementation details, and experimental results over the public remote sensing object datasets.

A. Datasets Description

In this section, we first introduce the datasets for the set of verifying experiments. Then, several groups of experiments have been implemented on these datasets, which are aimed to investigate the performance of the proposed method with comparisons to some state-of-the-art methods. All experiments are executed on a computer with a single Intel Core i7 CPU, an NVIDIA Titan X GPU, and 64GB of memory.

1) DOTA Dataset [48]: To evaluate the proposed approach, we first perform experiments on the DOTA dataset, which is the largest object detection dataset in the remote sensing community. It consists of 2806 remote sensing images from different sensors and platforms (e.g., Google Earth) with multiple spatial resolutions. The size of these image ranges from 800 $\times$ 800 pixels to 4000 $\times$ 4000 pixels. The DOTA dataset contains 15-classes objects including airplane, ship, baseball diamond (BD), bridge, ground field track (GFT), small vehicle (SV), large vehicle (LV), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), swimming pool (SP), harbor, and helicopter (HC).

The DOTA dataset randomly selected half of the original images as the training set, 1/6 as the validation set, and 1/3 as the testing set. In the process of experimental verifications, we adopt this default allocation ratio for a fair comparison.

2) NWPU VHR-10 Dataset [49]: To further verify the robustness of the proposed approach, we also implement experiments on the NWPU VHR-10 dataset, which contains a ten-class geospatial object detection dataset. The object classes of NWPU VHR-10 including airplane, ship, ST, BD, TC, BC, ground track field (GTF), harbor, bridge, and vehicle. This NWPU VHR-10 dataset consists of a positive image set including 650 images and a negative image set of 150 images over ten-class geospatial objects. These images are collected from Google Earth and Vaihigen dataset whose spatial resolutions are from 0.5 to 2.0 m and the image sizes of this dataset are range from 533 $\times$ 597 pixels to 1720 $\times$ 1028 pixels.

To ensure the standardization of this dataset, the experts in [29] crop the positive image set into a unified image size of 400 $\times$ 400 pixels, and acquire 1172 images. This modified dataset still contains the same ten-classes geospatial objects,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

and is manually annotated by experts. In this experiment, we randomly select 75% of images in the modified dataset as the training set, and the rest 25% of images in the modified dataset as the testing set.

3) HRRSD Dataset [8]: HRRSD is a large-scale HSRI object detection dataset, whose images are obtained from Google Earth, Baidu Map images. This dataset is composed of 21 761 color images acquired from Google Earth with the spatial resolution from 0.15 to 12 m, and 4961 color images collected from Baidu Map with the spatial solution from 0.6 to 1.2 m. There are 13-classes geospatial objects contained within the HRRSD dataset, i.e., airplane, ship, bridge, GTF, ST, BC, TC, BD, harbor, vehicle, crossroad (CR), T-junction (T-j), and parking lot (PL). The HRRSD consists of 55 740 object instances which are manually annotated by the experts in [8].

To guarantee the categories of object more balance, the whole dataset is split into three subsets, including the training set, the testing set, and the validation set. Each of the subsets maintains a relatively balanced image number and categories annotated object instances. Among the whole HRRSD dataset, 5401 images are selected as the training set, 10 943 images are selected as the testing set, and 5417 images are selected as the validation set. In this experiment, we adopt the same sample selection ratio for training and testing our proposed method.

B. Evaluation Metrics and Parameters Setting

To quantitatively evaluate the proposed method and other geospatial object detection methods, we utilize the precision–recall curve (PRC) and average precision (AP), which are standard and commonly adopted evaluation measures.

1) Precision–Recall Curve: The precision indicator measures the proportion of the detected object are true positives (TPs). The recall indicator measures the proportion of positives that are correctly recognized. TP, false positive (FP), and false negative (FN) denote the number of TP samples, the number of FP samples, and the number of FN samples, respectively. The precision and recall indicators are formulated as follows:

$$\text{Precison} = \frac{\text{TP}}{(\text{TP} + \text{FP})} \tag{19}$$

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FN})}. \tag{20}$$

Here, the detection is identified as TP if the intersection-over-union (IOU) between the predicted bounding box and the ground-truth bounding box exceeds 0.5; otherwise, the detection is considered as FP. In addition, if several detections overlap with the same ground-truth bounding box, only one is considered as TP, and the others are distinguished as FNs.

2) Average Precision: The AP refers to the area under the PRC, which is a comprehensive indicator that combines Precision and Recall. Meanwhile, the mean AP (mAP) is the mean of different category AP values and is the most important indicator in evaluating multicategory object detection. The larger of these two indicators are, the better of the performance of the detection algorithm, and vice versa. The formulation of mAP can be defined as

$$\text{mAP} = \frac{1}{n} \sum_{i}^{n} \text{AP}_i \tag{21}$$

where n is the number of object categories, and i is the category label. AP_i is the average precision of label i which denotes the integral area under the PRC.

C. Implementation Details and Parameter Optimization on Different Datasets

In our experiments, the architecture of the proposed EGAT-LSTM and hyperparameters are set cross validation. In order to achieve the best performance of our method, we empirically partition the neighbor set of each node into $K = 3$ subsets. Meanwhile, we design a distance-based partitioning strategy according to the average distance of each root node to its neighbor nodes. The average distance-based partitioning function can be formulated as

$$\ell_i(v_j) = \begin{cases} 0, & \text{if } p_{\text{avg}} = p(v_j, v_i) \\ 1, & \text{if } p_{\text{avg}} < p(v_j, v_i) \\ 2, & \text{if } p_{\text{avg}} > p(v_j, v_i) \end{cases} \tag{22}$$

where p_{avg} denotes the average distance from the root node to the neighbor nodes, and $p(v_j, v_i)$ is the distance between each node v_j to the root node v_i.

For the network and parameters setting, we utilize three LSTM layers for the main network, and one LSTM layer for each attention network, the channels of the three EGAT-LSTM layers are set to 512, and each LSTM layer composed of 100 LSTM neurons. During the process of training the EGAT-LSTM, the stochastic gradient descent (SGD) is applied as the optimization strategy, and the Nesterov momentum is set to 0.7. Droptout with a probability of 0.5 is adopted to avoid over-fitting on the three experimental datasets. Fig. 4 shows the training and validation loss curves of our proposed model on the three datasets, respectively.

We empirically find out that the learning rate γ, and the two-weight decaying coefficients of the regularizations μ and η are the three important parameters. The optimized combinations of these parameters play a significant effect for the object detection performance. Therefore, we first investigate how the detection performance is affected by these parameters on the validation sets of the DOTA, NWPU VHR-10, and HRRSD datasets, respectively. Specifically, the initial learning rates of them are set to $\gamma = \{0.005,\ 0.01\}$ which are reduced by multiplying them by 0.1 every 50 epochs, training stopped after 150 000 epochs or once there was no improvement on the validation set for more than 50 epochs. The two-weight decaying parameters are set to $\mu = \{0.00005, 0.0001, 0.0005, 0.001, 0.005\}$ and $\eta = \{0.0001, 0.0005, 0.001, 0.01, 0.05\}$, respectively.

As shown in Fig. 5, different combinations of these parameters govern the multiclass object detection results. Fig. 6 shows the multiclass object results implemented by our proposed EGAT-LSTM model on the three datasets. These results are measured in terms of mAP over all the object categories in the condition of different combinations of parameter settings.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

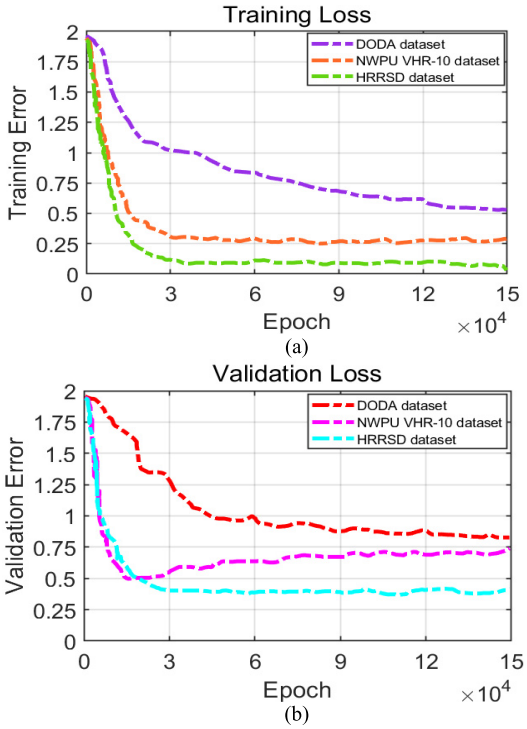


Fig. 5. Training and validation Errors on three public datasets. (a) Training errors on three public datasets. (b) Validation errors on three public datasets.

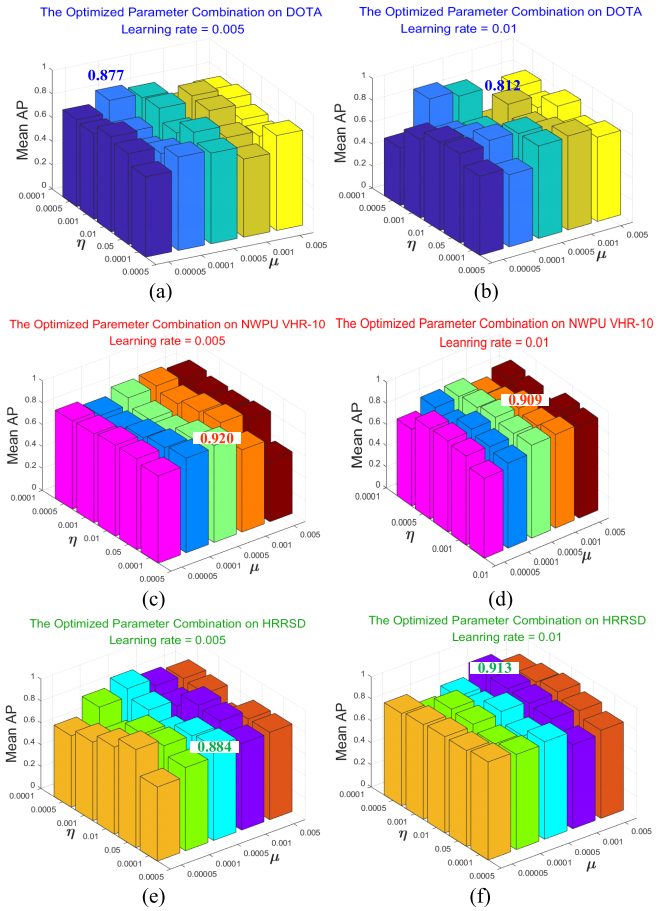


Fig. 6. Object detection results, measured based on the mAP over all object categories on different datasets, under different parameters of combinations. (a) Learning rate $\gamma = 0.005$ on DOTA. (b) Learning rate $\gamma = 0.01$ on DOTA. (c) Learning rate $\gamma = 0.005$ on NWPU VHR-10. (d) Learning rate $\gamma = 0.01$ on NWPU VHR-10. (e) Learning rate $\gamma = 0.005$ on HRRSD. (f) Learning rate $\gamma = 0.01$ on HRRSD.

Concretely, the best results on the DOTA dataset are achieved with $\gamma = 0.005$, $\eta = 0.001$, and $\mu = 0.001$, the highest performance on the NWPU VHR-10 is obtained with the parameter combination of $\gamma = 0.01$, $\eta = 0.0005$, and $\mu = 0.001$, and the greatest detection result on the HRRSD is acquired by the combination of $\gamma = 0.005$, $\eta = 0.0001$, and $\mu = 0.0005$, respectively. Therefore, we empirically set the above three optimized combination of these parameters in our subsequent experimental verifications on the three experimental datasets.

D. Experimental Results and Quantitative Comparisons on DOTA Dataset

1) Comparison of Each Module: To understand the contribution of each module used in the proposed EGAT-LSTM network, we first evaluate the object detection performance of the network on DOTA dataset by only using the spatial GCNs module (Sp-G) and semantic GCNs module (Se-G), respectively, as well as adopting different combinations of them with the attention module (Atten) under the Siamese structure (Siam). The detailed network configurations and their corresponding experimental results are summarized in Table II. As shown in Table II, the mAPs of Sp-G + Atten and Se-G + Atten are improved by 2.9% and 3.29% compared to the mAPs of Sp-G and Se-G, respectively. This indicates the effectiveness of the attention module for enhancing the ability for spatial or semantic graph relations reasoning. After integrating with the Siamese structure (Sp-G + Se-G + Siam) for spatial and semantic relations discrimination, the improvements of 7.62% and 6.25% are can be achieved, which are compared to the mAPs obtained by the Sp-G + Atten and Se-G + Atten, respectively. This validates that the Siamese structure plays an important role in improving the detection performance. In addition, by applying the above network modules for learning the proposed EGAT-LSTM network (Sp-G + Se-G + Siam + Atten), we can achieve an improvement of 6.25% in term of mAP. This demonstrates that the combinations of these submodules can effectively learning the complete spatial–semantic graph features for relations reasoning, and also leading to better performance for multiclass geospatial object detection in HRSI.

2) Comparison With Other State-of-the-Art Methods: To further evaluate the effectiveness of the proposed EGAT-LSTM for multiclass geospatial object detection, we implement it on the DOTA dataset, and then compare it with six state-of-the-art detection methods. We conduct several experiments on the large-scale dataset with the same default data split ratio setting as the DOTA dataset. These compared methods consist of Faster R-CNN [41], region-based fully convolutional networks (R-FCNS) [50], single-shot multibox detector (SSD) [51], multiscale spatial and channel-wise attention model (MSCA) [18], spatial temporal graph convolutional net-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

TABLE II
PERFORMANCE OF MULTICLASS OBJECT DETECTION ON DOTA DATASET UNDER DIFFERENT CONFIGURATIONS OF EGAT-LSTM IN TERMS OF AP VALUES (%)

	Network Configurations				airplane	ship	BD	bridge	GTF	SV	LV	TC	BC	ST	SBF	RA	SP	harbor	HC	mAP
	Sp-G	*Sm-G*	*Siam*	*Atten*																
EGAT-LSTM	✓	-	-	-	88.42	82.42	73.73	51.81	61.93	66.21	79.59	87.35	81.93	79.45	55.31	59.97	65.97	75.45	68.12	71.91
	-	✓	-	-	88.96	82.56	74.31	53.92	62.13	66.62	81.36	86.67	82.92	80.94	59.62	62.39	66.56	75.25	69.25	72.89
	✓	-	-	✓	89.54	82.42	77.23	53.38	66.72	69.03	79.93	90.78	86.39	85.32	63.92	65.61	69.76	73.24	67.92	74.81
	-	✓	-	✓	89.13	82.52	83.21	55.73	73.03	70.63	78.86	89.07	86.24	85.21	65.56	67.94	75.52	73.95	67.24	76.18
	✓	✓	✓	-	93.85	88.56	87.56	67.97	78.45	80.24	82.98	94.24	92.79	90.18	69.21	75.97	82.34	79.26	72.87	82.43
	✓	✓	✓	✓	96.63	90.89	88.93	86.32	86.12	85.42	91.96	95.63	89.87	92.12	86.19	84.11	81.96	86.45	74.23	87.78

work (ST-GCN) [52], and you only look once (YOLOv2) [53]. Fig. 7 demonstrates visualized detection results of 15-classes objects of DOTA dataset using the proposed EGAT-LSTM.

As can be seen from Fig. 7, our method indicates a reasonable detection performance on the classes of airplane, ship, SP, BD, and harbor. In Fig. 7(a)–(d), for the airplanes, HC and SP with different sizes and different directions are distributed in a long-range spatial locations, the EGAT-LSTM shows a perfect detection performance. In Fig. 7(e)–(h), the spatial distributions of object categories are more complex such as the ships, ST, LV, BC, and BD, they also layout compactly and have high appearance ambiguity. In this situation, the EGAT-LSTM also exhibits an almost perfect detection performance. The corresponding quantitative comparison results are shown in Table III. As can been seen from Table III, for most object categories, the proposed EGAT-LSTM makes a considerable promotion compared with classic CNN or GCN frameworks different approaches for all 15-classes geospatial objects in terms of the mAP. The mAP value is improved from 10.94% of ST-GCN to 48.58% of YOLOv2. Specifically, our proposed EGAT-LSTM achieves an improvement of 10.94%, 27.14%, 30.64%, 25.77%, 38.62%, and 48.58%, compared with the ST-GCN, Faster-RCNN, R-FCN, MSCA, SSD, and YOLOv2, respectively. These experimental results demonstrate the superiority of the proposed EGAT-LSTM compared with the other six state-of-the-art methods.

3) Comparison With Different LSTM Mechanism-Based Fused Features: Moreover, consider that our method mainly takes advantage of the LSTM mechanism to efficiently fuse spatial and semantic graph-structured features for improving the performance of object detection. Therefore, we also compare the spatial-based relation graph features, the semantic-based relation graph features, and the embedding spatial–semantic relations features, respectively, for multiclass geospatial objects detection. Here, we note that the spatial and semantic graphs can be regarded as the explicit visual relations of objects–objects, which are evolved from the object proposal region-based features acquired by the region-based CNN visual feature extractor. Based on these two types of explicit relations graphs, we draw inspiration from recent advances of the GCNs-based visual relations reasoning, and integrate it with the LSTM-mechanism for further deducing implicit semantic-spatial relations graphs, which is aimed to improve the performance of object detection.

Specifically, we adopt the average errors of APs over the 15-classes to evaluate the detection results which are obtained with different features, the compared results are shown in Fig. 8. In terms of the mAP, the detected performance using the embedded spatial–semantic graph features improves spatial graph features by 20.10%, and improves semantic graph features by 14.29%, respectively. It is worth mentioning that the LSTM-based mechanism features fusion strategy achieves significant performance improvements for multiclass geospatial object detection HRSI.

4) Cross-Validation Using Different Split Ratios of DOTA: As we have known, the DOTA dataset exists the sample-imbalanced phenomenon, and different kinds of object instances within the images exhibit diverse spatial distribution scales. Meanwhile, they also present large-scale variations of appearance attributes between different categories of objects. To verify the robustness and effectiveness of the proposed EGAT-LSTM, we randomly adopt different data split ratios of this dataset for investigating the object detection performance of our method. Concretely, we utilize ten split ratios [train (%)**:**val(%)**:**test(%)] to divide the original DOTA dataset into the training, validation and testing sets. The detailed experimental results under the corresponding split strategy are summarized in Table IV. As can been seen from Table IV, the mAPs using the EGAT-LSTM improve quickly from the split ratio of 20:40:60 to the split ratio of 50:20:30. The mAP gains are 71.97%, 73.05%, 76.68%, 78.70%, and 83.62% in the above range of split ratios. As the ratio of training set is larger than 50%, the mAP values achieved by the following ratios of the EGAT-LSTM all exceed 83.62%, and then they are all remaining at a relatively high mAP value. Using the ratio of 80:20:20 achieves the best object performance among all the experimental results, gaining an mAP value of 88.78%. These results demonstrate the robustness and the effectiveness of our proposed EGAT-LSTM for multiclass object detection.

E. Experimental Results and Quantitative Comparisons on NWPU VHR-10 Dataset

1) Comparison of Each Module: To evaluate the efficiency of each module used in the proposed EGAT-LSTM network, we conduct another group of object detection experiments on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

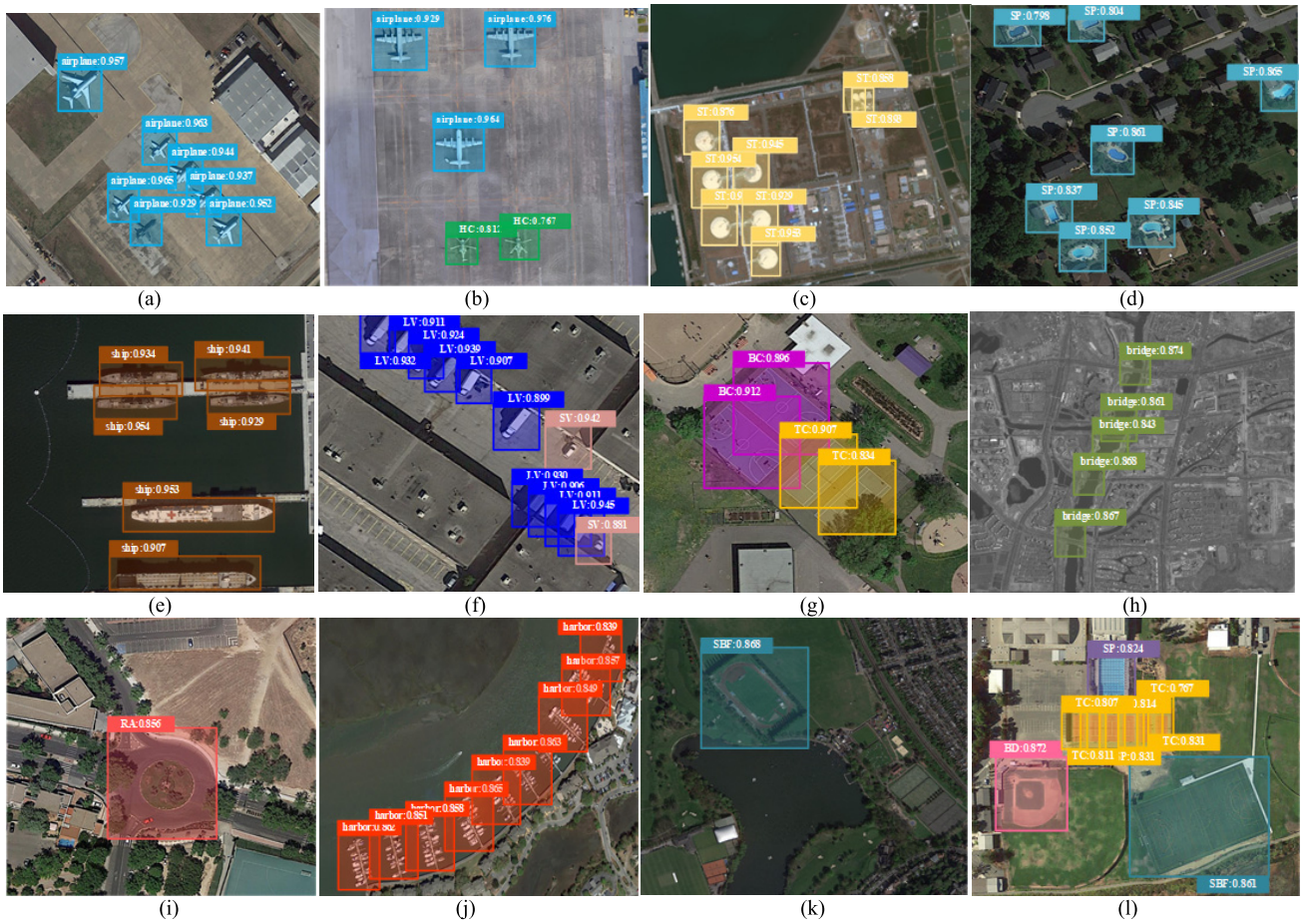


Fig. 7. Multiclass object detection results of the proposed method on the DOTA dataset. (a) Airplane. (b) Airplane and HC. (c) ST. (d) SP. (e) Ship. (f) LV and SV. (g) BC and TC. (h) Bridge. (i) RA. (j) Harbor. (k) SBF. (l) SP, TC SBF, and BD.

TABLE III
PERFORMANCE COMPARISONS OF SEVEN DIFFERENT APPROACHES ON DOTA DATASET IN TERMS OF AP VALUES (%)

Method	airplane	ship	BD	bridge	GTF	SV	LV	TC	BC	ST	SBF	RA	SP	harbor	HC	mAP
YOLOv2 [53]	76.90	52.37	33.87	22.73	34.88	38.73	32.02	61.65	48.54	33.91	29.27	36.83	38.26	36.44	11.61	39.20
SSD [51]	76.84	41.70	57.26	27.09	30.73	23.96	56.91	80.95	37.29	36.01	33.80	32.98	29.57	63.19	20.46	43.25
MSCA [18]	78.11	48.27	67.68	28.30	42.16	23.99	62.24	82.64	44.96	38.37	40.46	36.37	38.50	69.21	36.10	49.16
R-FCN [50]	86.95	77.20	62.19	42.11	50.86	42.32	74.59	84.71	53.03	60.70	43.17	42.73	46.91	64.91	24.64	57.14
Faster-RCNN [41]	80.32	50.04	77.55	32.86	68.13	53.66	52.49	90.41	75.05	59.59	57.00	49.81	56.46	61.69	41.85	60.64
ST-GCN [52]	90.42	86.42	79.87	63.39	76.54	77.46	87.87	86.83	80.08	82.45	68.19	69.43	65.08	81.41	57.17	76.84
Our method	96.63	90.89	88.93	86.32	86.12	85.42	91.96	95.63	89.87	92.12	86.19	84.11	81.96	86.45	74.23	87.78

NWPU VHR-10 dataset, which are also implemented by only using the Sp-G and Se-G, respectively, as well as adopting different combinations of them with the attention module (Atten) under the Siamese structure (Siam). The detailed network configurations and their corresponding experimental results are summarized in Table V. As shown in Table V, the mAPs of Sp-G + Atten and Se-G + Atten are improved by 3.38% and 3.26% compared to the mAPs of Sp-G and Se-G, respectively. After integrating with the modules of Sp-G and Se-G under Siamese structure (Sp-G + Se-G + Siam), the improvements of 5.22% and 2.98% are can be achieved by our method, which are compared to the mAPs obtained under the configurations of Sp-G + Atten and Se-G + Atten, respectively. By applying the network configuration of Sp-G + Se-G + Siam + Atten, we acquire an improvement of 3.25% in term of mAP. This demonstrates that fusing multiple submodules for spatial and semantic relations reasoning can capture more complete relations of objects–objects, which helps to diversity object spatial relations and enhance semantic information for forming powerful features, and then can improve the performance for multiclass object detection.

2) Comparison With Other State-of-the-Art Methods: To verify the effectiveness of the proposed method, we also produce a set of experiments on the NWPU VHR-10 dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

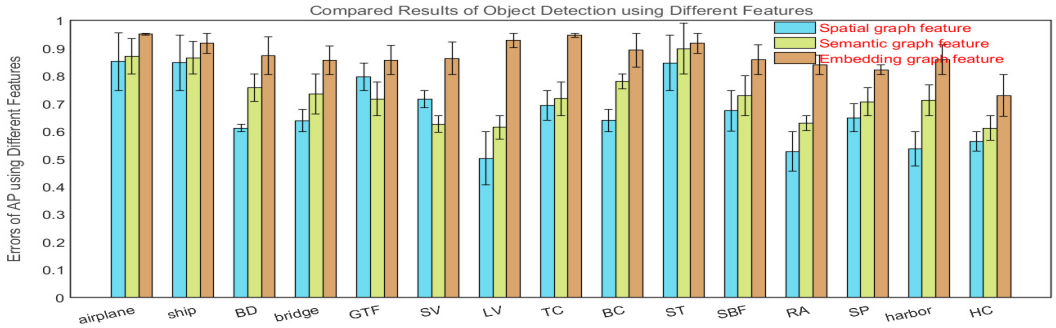


Fig. 8. Multiclass object detection results using different features of our proposed EGAN-LSTM framework on the DOTA dataset.

TABLE IV

PERFORMANCE OF MULTICLASS OBJECT DETECTION ON DOTA DATASET UNDER DIFFERENT SPLIT RATIOS OF EGAT-LSTM IN TERMS OF AP VALUES (%)

Data Split Ratio *train(%):val(%):test(%)*	airplane	ship	BD	bridge	GTF	SV	LV	TC	BC	ST	SBF	RA	SP	harbor	HC	mAP
20:40:60	86.13	83.02	73.45	53.91	62.44	64.21	80.17	84.66	82.43	80.79	57.92	61.49	65.34	75.98	67.71	71.97
20:60:20	87.67	83.22	75.19	54.17	62.25	65.97	82.79	85.97	84.14	81.33	58.27	63.11	67.24	76.08	68.38	73.05
40:40:20	89.12	84.21	79.27	62.14	76.55	77.31	87.03	87.29	84.07	81.23	64.79	66.47	62.94	81.39	66.45	76.68
40:20:40	90.05	85.97	82.04	64.79	78.92	78.87	87.24	90.02	86.73	83.45	66.77	68.12	64.23	84.17	69.25	78.70
50:20:30	93.97	89.42	88.76	70.07	79.93	81.78	84.02	94.11	94.15	91.87	70.29	76.78	84.29	80.96	74.01	83.62
50:30:20	94.78	90.65	88.98	72.33	81.16	82.07	85.24	95.71	94.37	93.29	71.78	77.65	84.31	82.89	76.01	84.74
60:20:20	95.29	92.78	89.04	75.21	82.89	84.56	86.43	97.22	93.12	95.32	73.49	78.19	86.98	83.57	78.87	86.19
70:20:10	95.43	93.92	90.16	75.54	84.12	84.61	86.87	97.51	95.01	96.89	75.03	79.18	87.95	84.28	79.82	87.08
70:10:20	96.03	93.46	90.34	76.51	84.15	85.72	86.98	98.03	95.26	97.11	75.67	80.18	88.19	85.76	80.54	87.59
80:20:20	96.45	93.77	91.01	76.53	85.16	85.98	97.02	98.17	96.23	97.67	76.42	80.32	88.97	86.56	81.29	88.77

TABLE V

PERFORMANCE OF OBJECT DETECTION ON NWPU VHR-10 DATASET UNDER DIFFERENT CONFIGURATIONS OF EGAT-LSTM IN TERMS OF AP VALUES (%)

	Network Configurations				airplane	ship	bridge	GTF	ST	BC	TC	BD	harbor	vehicle	mAP
	Sp-G	*Sm-G*	*Siam*	*Atten*											
EGAT-LSTM	✓	-	-	-	92.51	80.65	62.19	94.76	90.03	60.16	81.13	86.97	81.08	72.18	80.16
	-	✓	-	-	94.87	83.65	66.56	95.19	92.39	62.76	83.42	87.03	83.81	75.61	82.52
	✓	-	-	✓	95.61	84.98	67.92	94.78	93.76	64.57	84.21	88.91	84.56	76.16	83.54
	-	✓	-	✓	96.98	86.19	73.12	96.51	95.78	67.34	86.97	90.94	87.11	76.87	85.78
	✓	✓	✓	-	97.76	89.85	79.31	96.85	96.34	71.03	87.98	92.67	88.89	80.08	88.76
	✓	✓	✓	✓	97.32	96.72	80.14	94.18	97.16	94.46	86.64	96.51	86.18	90.79	92.01

with the same data setting, for a fair comparison of multiclass object detection. Then, we also compare these experimental results with the above six state-of-the-art methods which are adopted on the DOTA dataset. These contrast methods include the Faster R-CNN [41], R-FCN [50], SSD [51], MSCA [18], ST-GCN [52], and YOLOv2 [53]. Fig. 9 shows the result of each class of the object on NWPU VHR-10 dataset with our method. The proposed EGAT-LSTM achieves a good performance on ten-class geospatial objects. For a fair comparison, we note that all the comparative experiments are adopted the same data setting as the default setting on the NWPU VHR-10 dataset.

Table VI and Fig. 10 indicate the quantitative comparison results. As can be seen from them, the proposed approach exhibits a better performance compared with other different approaches for all ten-class geospatial objects in terms of AP. Concretely, our method achieves the performance improvements of 3.51%, 6.93%, 11.11%, 10.81%, 13.61%, 15.21%, and 15.71%, in terms of the mAP over ten-class object categories, compared with the ST-GCN, MSCA, faster R-CNN, SSD, YOLOv2, and R-FCN, respectively. As shown in Table VI, for some object categories with complex spatial distribution attributes, our method obtains high significant improvement, especially for the object classes of the airplane, ship, vehicle, and storage. Moreover, for some geospatial object classes with high appearance ambiguity characteristics, such as the BD, TC, and BC, the proposed method also can achieve the best detection performance. These quantitative results demonstrate the superiority and robustness of our proposed method compared with the other six state-of-the-art methods.

F. Experimental Results and Quantitative Comparisons on HRRSD Dataset

Compared with the datsets of DOTA and NWPU VHR-10, the HRRSD dataset has the more reasonable data configurations including the balanced subsets, object categories, and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

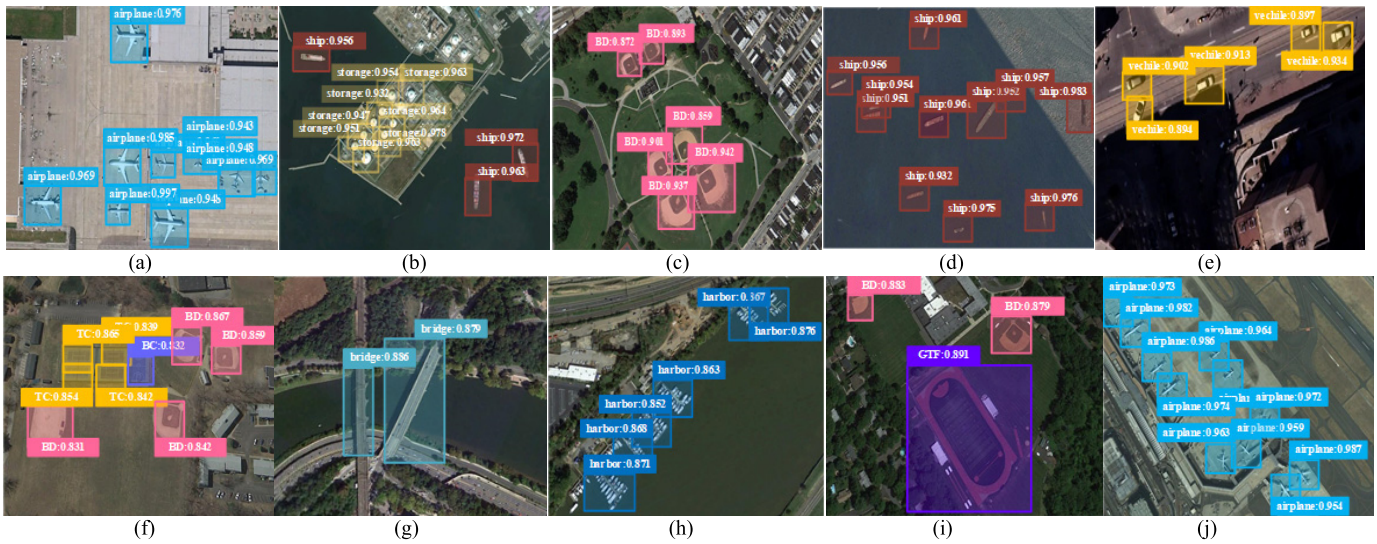


Fig. 9. Multiclass object detection results of the proposed method on the NWPU VHR-10 dataset. (a) Airplane. (b) Ship and storage. (c) BD. (d) Ship. (e) Vehicle. (f) TC, BD, and BCC. (g) Bridge. (h) Harbor. (i) GTF and BD. (j) Airplane.

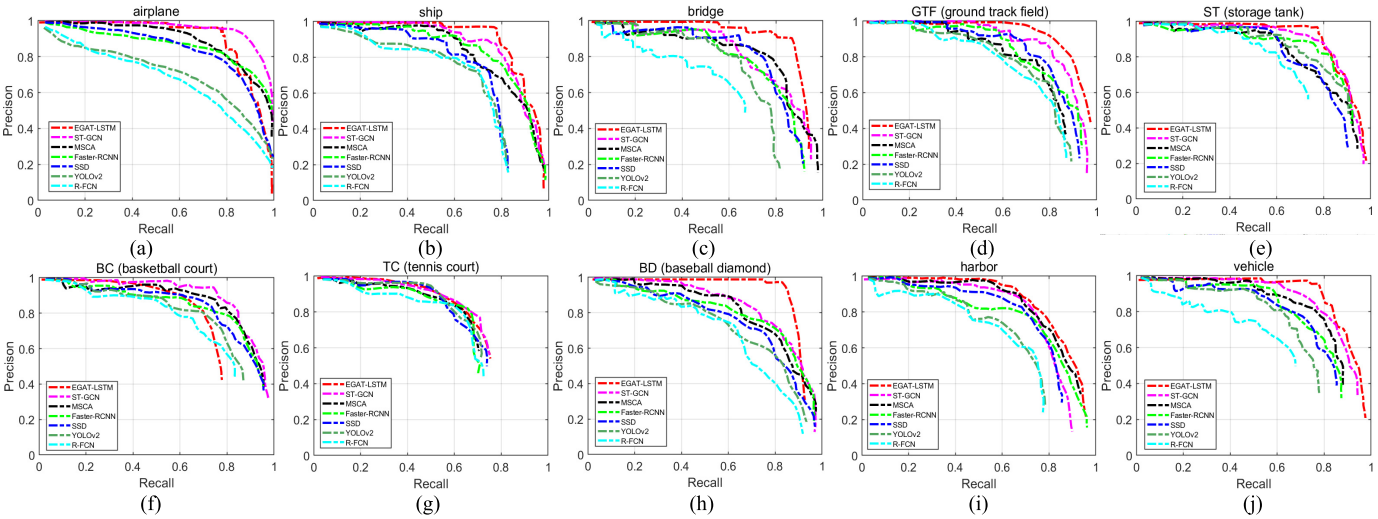


Fig. 10. PRCs of the compared method and ours proposed method on the NWPU VHR-10 dataset. (a) Airplane. (b) Ship. (c) Bridge. (d) GTF. (e) ST. (f) BC. (g) TC. (h) BD. (i) Harbor. (e) Vehicle.

TABLE VI

PERFORMANCE COMPARISONS OF EIGHT DIFFERENT APPROACHES ON NWPU VHR-10 DATASET IN TERMS OF AP VALUES (%)

Method	airplane	ship	bridge	GTF	ST	BC	TC	BD	harbor	vehicle	mAP
R-FCN [50]	81.70	80.60	47.80	89.80	66.20	69.70	80.20	90.30	78.60	78.30	76.30
YOLOv2 [53]	83.00	84.40	85.00	62.80	81.90	53.50	85.00	84.30	78.70	70.00	76.80
SSD [51]	90.40	60.90	76.70	98.30	79.80	80.60	82.60	89.90	73.40	52.10	78.40
Faster-RCNN [41]	94.60	82.30	57.50	92.40	65.32	89.70	81.90	95.50	72.40	77.80	80.90
MSCA [18]	99.47	79.95	65.85	100	90.39	77.33	90.55	90.52	76.10	80.61	85.08
ST-GCN [52]	95.76	94.82	73.41	87.62	95.03	86.92	85.83	93.45	84.86	87.39	88.50
Our method	97.32	96.72	80.14	94.18	97.16	94.46	86.64	96.51	86.18	90.79	92.01

number of object instances that is evenly distributed in each object category. In order to further investigate the generality and the robustness of our method, we have constructed another group of contrastive experiments with different state-of-the-art methods, including Transform-CNN [54], RICNN [29], YOLOv2 [53], gated axis-concentrated localization network faster-RCNN (GACL Faster-RCNN) [55], and multiple graph convolutional network (MGCN) [43]. Compared with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

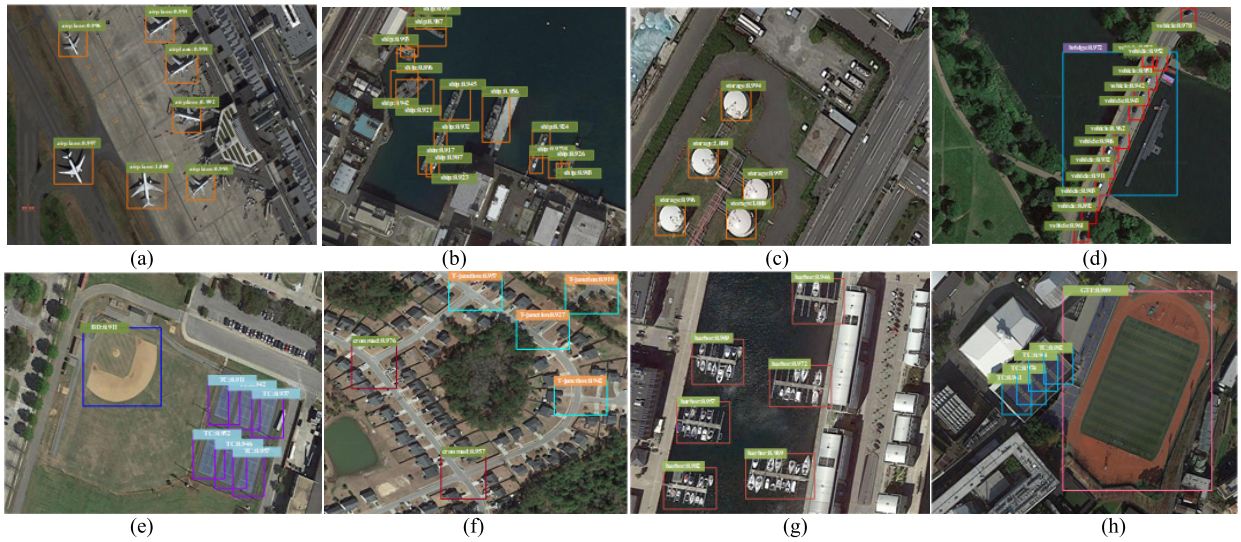


Fig. 11. Multiclass object detection results of the proposed method on the HRRSD dataset. (a) Airplane. (b) Ship. (c) Storage. (d) Bridge and vehicle. (e) Court and BD. (f) CR and T-j. (g) Harbor. (h) TC and GTF.

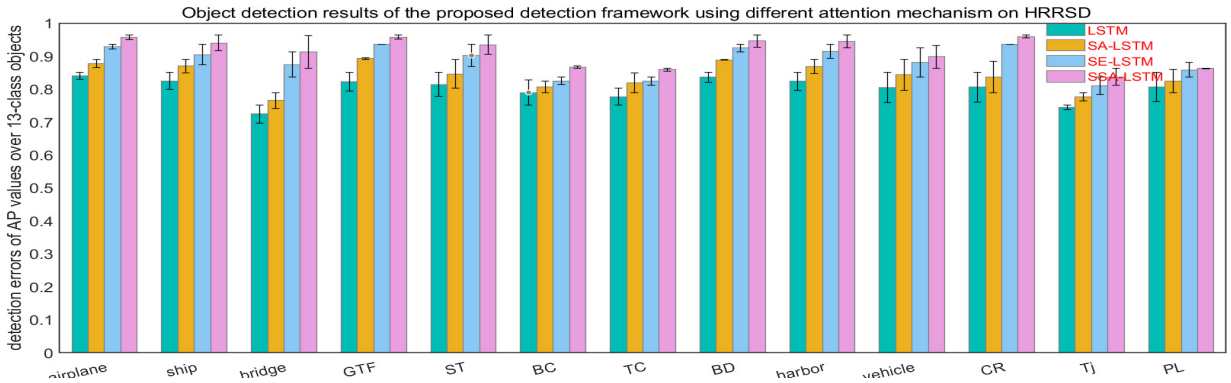


Fig. 12. Multiclass object detection results using different attention mechanism of our proposed EGAN-LSTM framework on the HRRSD dataset.

previous datasets, the HRRSD contains more balance object categories and enough samples for training the network. However, each image of this dataset consists of greater number of objects and richer object categories, while these multiclass geospatial objects also exhibit a more diverse range of spatial distribution and larger appearance and semantic ambiguity. These factors also pose the more challenges for object detection. Fig. 11 shows the visualized results over the 13-classes geospatial object detection using the proposed method.

Table VII and Fig. 12 list the corresponding quantitative results in terms of AP. Clearly, our proposed EGAT-LSTM model achieves the best performance among all the methods, with an improvement of 45.98% compared to transform-CNN, 43.15% compared to RICNN, 25.50% compared to YOLOv2, 9.2% compared to GACL Faster-RCNN, and 5.03% compared to MGCN. Thereupon, we can make the following analysis. It is obvious that transform-CNN and RICNN achieve the worst performances because they all ignore the spatial relationships among different kinds of geospatial objects, as well as limited by their spatial discrimination ability of multiclass object distributions. Although the YOLO2 can yield better performance than transform-CNN and RICNN by modeling a deeper network and embedding anchor boxes, it is not robust for small scale objects and arbitrary pairs of objects that are not more than a tiny distance apart. The GACL faster-RCNN implements a channel attention mechanism-based on local feature learning that achieves a huge improvement for detection performance. Compared with the above four methods, MGCN holds a slightly lower performance than our method but much better than others. It is because the MGCN only focuses on the multiple spatial scale for graph-structured features learning and ignores the reasoning for semantic relationships between different pairs of constructed graphs. As expected, the detection performance of the proposed EGAT-LSTM largely outperforms the other investigated methods on the HRRSD dataset over 13-class object categories, which shows its effectiveness and superiority.

The main purpose of the proposed EGAT-LSTM is to explore the explicit and implicit visual relations for improving the object detection performance. It is implemented by integrating with the proposed embedded LSTM-based attention mechanism, and the spatial and semantic graph-structured features learning from the GCNs. Specifically, the spatial and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
PERFORMANCE COMPARISONS OF SIX DIFFERENT APPROACHES ON HRRSD DATASET IN TERMS OF AP VALUES (%)

Method	airplane	ship	bridge	GTF	ST	BC	TC	BD	harbor	vehicle	CR	Tj	PL	mAP
Transform-CNN [54]	77.46	49.68	19.98	76.28	79.10	17.97	70.83	57.60	54.13	41.27	25.90	2.42	16.57	45.32
RICNN [29]	78.05	56.54	27.44	78.00	80.95	22.95	66.37	59.64	47.81	51.95	26.63	9.33	24.45	48.15
YOLOv2 [53]	84.60	78.50	79.00	94.40	72.40	41.30	67.60	62.20	74.40	65.10	43.40	46.80	45.80	65.80
GACL Faster-RCNN [55]	90.80	87.20	85.60	90.70	89.20	49.70	80.80	87.20	89.70	86.90	88.20	75.00	65.30	82.10
MGCN [43]	93.27	88.96	89.79	93.46	92.10	55.79	83.98	91.73	92.96	89.85	94.95	79.87	74.69	86.27
Our method	95.67	93.92	91.23	95.72	93.45	86.63	85.89	94.52	94.36	89.76	95.86	83.71	86.22	91.30

TABLE VIII
PERFORMANCE COMPARISONS OF OUR EGAT-LSTM USING DIFFERENT ATTENTION MECHANISMS ON HRRSD DATASET OVER 13-CLASS OBJECTS IN TERMS OF AP VALUES (%)

Object Class	LSTM	SA-LSTM	SE-LSTM	SSA-LSTM
airplane	83.98	87.68	92.86	95.67
ship	82.45	86.92	90.42	93.92
bridge	72.39	76.54	87.35	91.23
GTF	82.21	89.23	93.51	95.72
ST	81.39	84.57	90.15	93.45
BC	78.93	80.68	82.42	86.63
TC	77.67	81.92	82.38	85.89
BD	83.59	88.84	92.39	94.52
harbor	82.34	86.78	91.39	94.36
vehicle	80.46	84.25	87.97	89.76
CR	80.56	83.64	93.57	95.86
Tj	74.42	77.68	80.89	83.71
PL	80.62	82.39	85.78	86.22
mAP	80.07	83.93	88.54	91.30

semantic relations deducing from the spatial and semantic LSTM-based module respectively, are regarded as the explicit visual relations. Meanwhile, the embedded spatial–semantic relations can be defined as the implicit visual relations. Therefore, to validate the effectiveness of our designed spatial–semantic-based attention mechanism, we conduct another group of experiments for object detection, which are implemented by adopting different LSTM-based configurations on the HRRSD dataset. These configurations consist of four attention-based fusion strategies.

1) *LSTM*: The main LSTM network without other attention design in the subnets (LSTM).

2) *SA-LSTM*: The combination of LSTM in the main network and spatial attention (SA) in the Sp-G s.

3) *SE-LSTM*: The combination of LSTM in the main network and semantic attention in the Se-G.

4) *SSA-LSTM*: The combination of LSTM in the main network and the embedding SSA GCNs.

The quantitative comparison results are shown in Table VIII and Fig. 11. As can be seen from Table VIII, our designed SSA-LSTM model achieves the best performance among all the compared attention fused configurations. Specially, compared with the baseline scheme in the main network, in terms of the mAP values of detection results over 13-class objects, the introduction of the SA-LSTM module and the SE-LSTM brings up to 3.86% and 8.47%, respectively. The best performance is achieved by our designed SSA-LSTM, with the improvement of 11.23%. Fig. 11 demonstrates the average errors in terms of mAP values over 13-class objects detection results, in which the smoothness of the errors variations also indicates the highest robustness of our devised SSA-LSTM model. These comparison results prove that the introduction of SSA-LSTM in the EGAT-LSTM is optimal for object detection compared with other methods.

V. CONCLUSION

In this article, we propose a novel EGAN-LSTM framework for multiclass geospatial object detection in HSRI. It parameterizes the graph-structured data which is conducted from the image data, and then embeds it into the network for jointly learning and optimizing the model. The EGAN-LSTM is composed of two attention mechanism-based modules under the Siamese architecture, including the spatial GAT and the semantic GAT. The proposed EGAT-LSTM can not only capture the discriminative appearance features in the semantic attention module but also explore the global and local spatial relations among different categories of objects. By this way, the learned features achieved from the jointly LSTM mechanism-based, can effectively address the problems of the spatial interference and the appearance ambiguity among different kinds of geospatial objects. Experiments that conducted on three public datasets demonstrate the superiority of our method compared with some publishing state-of-the-art methods. In the future work, we will focus on how to improve the flexibility and adaptivity of our EGAT-LSTM model, and also pay attention to the newly developed feature fusion strategy of further improving the performance of our model.

REFERENCES

[1] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[2] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[3] Y. Li, Y. Zhang, X. Huang, and A. L. Yuille, "Deep networks under scene-level supervision for multi-class geospatial object detection from remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 182–196, Dec. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

[4] M. Zhou, Z. Zou, Z. Shi, W.-J. Zeng, and J. Gui, "Local attention networks for occluded airplane detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 3, pp. 381–385, Mar. 2020.

[5] G. Cheng, J. Han, P. Zhou, and D. Xu, "Learning rotation-invariant and Fisher discriminative convolutional neural networks for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 265–278, Jan. 2019.

[6] Z. Zhang, W. Guo, S. Zhu, and W. Yu, "Toward arbitrary-oriented ship detection with rotated region proposal and discrimination networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1745–1749, Nov. 2018.

[7] W. Zhang, X. Sun, K. Fu, C. Wang, and H. Wang, "Object detection in high-resolution remote sensing images using rotation invariant parts based model," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 74–78, Jan. 2014.

[8] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.

[9] Y. Zhang, K. Fu, H. Sun, X. Sun, X. Zheng, and H. Wang, "A multi-model ensemble method based on convolutional neural networks for aircraft detection in large remote sensing images," *Remote Sens. Lett.*, vol. 9, no. 1, pp. 11–20, Jan. 2018.

[10] W. Diao, X. Sun, X. Zheng, F. Dou, H. Wang, and K. Fu, "Efficient saliency-based object detection in remote sensing images using deep belief networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 2, pp. 137–141, Feb. 2016.

[11] X. Han, Y. Zhong, and L. Zhang, "An efficient and robust integrated geospatial object detection framework for high spatial resolution remote sensing imagery," *Remote Sens.*, vol. 9, no. 7, p. 666, Jun. 2017.

[12] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2019.

[13] L. Mou, Y. Hua, and X. X. Zhu, "A relation-augmented fully convolutional network for semantic segmentation in aerial scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12416–12425.

[14] U. Chaudhuri, B. Banerjee, and A. Bhattacharya, "Siamese graph convolutional network for content based remote sensing image retrieval," *Comput. Vis. Image Understand.*, vol. 184, pp. 22–30, Jul. 2019.

[15] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[16] X. Wang and A. Gupta, "Videos as space-time region graphs," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 399–417.

[17] L. Wu, Y. Wang, X. Li, and J. Gao, "Deep attention-based spatially recursive networks for fine-grained visual recognition," *IEEE Trans. Cybern.*, vol. 49, no. 5, pp. 1791–1802, May 2019.

[18] J. Chen, L. Wan, J. Zhu, G. Xu, and M. Deng, "Multi-scale spatial and channel-wise attention for improving object detection in remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 4, pp. 681–685, Apr. 2020.

[19] X. Liang, X. Shen, D. Xiang, J. Feng, L. Lin, and S. Yan, "Semantic object parsing with local-global long short-term memory," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3185–3193.

[20] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997, doi: 10.1162/neco.1997.9.8.1735.

[21] B. Zhao, X. Li, and X. Lu, "CAM-RNN: Co-attention model based RNN for video captioning," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5552–5565, Nov. 2019.

[22] Q. Li, G. Wang, J. Liu, and S. Chen, "Robust scale-invariant feature matching for remote sensing image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 287–291, Apr. 2009.

[23] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, no. 1, Jun. 2005, pp. 886–893.

[24] D. Zhang, J. Han, G. Cheng, Z. Liu, S. Bu, and L. Guo, "Weakly supervised learning for target detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 701–705, Apr. 2015.

[25] N. Yokoya and A. Iwasaki, "Object detection based on sparse representation and Hough voting for optical remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2053–2062, May 2015.

[26] J. Han *et al.*, "Efficient, simultaneous detection of multi-class geospatial targets based on visual saliency modeling and discriminative learning of sparse coding," *ISPRS J. Photogramm. Remote Sens.*, vol. 89, pp. 37–48, Mar. 2014.

[27] G. E. Hinton, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, Jul. 2006.

[28] J. Han, D. Zhang, G. Cheng, N. Liu, and D. Xu, "Advanced deep-learning techniques for salient and category-specific object detection: A survey," *IEEE Signal Process. Mag.*, vol. 35, no. 1, pp. 84–100, Jan. 2018.

[29] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[30] D. Zhang, D. Meng, and J. Han, "Co-saliency detection via a self-paced multiple-instance learning framework," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 5, pp. 865–878, May 2017.

[31] Y. Zhu, R. Urtasun, R. Salakhutdinov, and S. Fidler, "SegDeepM: Exploiting segmentation and context in deep neural networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4703–4711.

[32] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[33] C.-Y. Fu, W. Liu, A. Ranga, A. Tyagi, and A. C. Berg, "DSSD : Deconvolutional single shot detector," 2017, *arXiv:1701.06659*. [Online]. Available: http://arxiv.org/abs/1701.06659

[34] S. Chen, B. Wang, X. Tan, and X. Hu, "Embedding attention and residual network for accurate salient object detection," *IEEE Trans. Cybern.*, vol. 50, no. 5, pp. 2050–2062, May 2020.

[35] J. Kuen, Z. Wang, and G. Wang, "Recurrent attentional networks for saliency detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3668–3677.

[36] C. Wang, X. Bai, S. Wang, J. Zhou, and P. Ren, "Multiscale visual attention networks for object detection in VHR remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 310–314, Feb. 2019.

[37] X. Ying *et al.*, "Multi-attention object detection model in remote sensing images based on multi-scale," *IEEE Access*, vol. 7, pp. 94508–94519, Jul. 2019.

[38] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Multisource region attention network for fine-grained object recognition in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4929–4937, Jul. 2019.

[39] Y. Chen, M. Rohrbach, Z. Yan, Y. Shuicheng, J. Feng, and Y. Kalantidis, "Graph-based global reasoning networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 433–442.

[40] N. Khan, U. Chaudhuri, B. Banerjee, and S. Chaudhuri, "Graph convolutional network for multi-label VHR remote sensing scene recognition," *Neurocomputing*, vol. 357, pp. 36–46, Sep. 2019.

[41] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[42] K. Marino, R. Salakhutdinov, and A. Gupta, "The more you know: Using knowledge graphs for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 21–26.

[43] B. Jiang, X. Jiang, J. Tang, B. Luo, and S. Huang, "Multiple graph convolutional networks for co-saliency detection," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2019, pp. 332–337.

[44] Z. Zhang, P. Cui, and W. Zhu, "Deep learning on graphs: A survey," 2018, *arXiv:1812.04202*. [Online]. Available: http://arxiv.org/abs/1812.04202

[45] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–12.

[46] C.-H. He, S.-C. Lai, and K.-M. Lam, "Improving object detection with relation graph inference," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2019, pp. 2537–2541.

[47] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[48] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[49] K. Li, G. Cheng, S. Bu, and X. You, "Rotation-insensitive and context-augmented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2337–2348, Apr. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.

[50] J. Dai, Y. Li, K. He, and J. Sun, "R-FCN: Object detection via region-based fully convolutional networks," in *Proc. Adv. Neural Inf. Process. Syst., Annu. Conf. Neural Inf. Process. Syst.*, Barcelona, Spain, Dec. 2016, pp. 379–387.
[51] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands, Oct. 2016, pp. 21–37.
[52] S. Yan, Y. Xiong, and D. Lin, "Spatial temporal graph convolutional networks for skeleton-based action recognition," in *Proc. 2nd AAAI Conf. Artif. Intell.*, 2018.
[53] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.
[54] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.
[55] X. Lu, Y. Zhang, Y. Yuan, and Y. Feng, "Gated and axis-concentrated localization network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 179–192, Jan. 2020.



Shu Tian received the B.Sc. degree in electronic information engineering from Jilin Engineering Normal University, Changchun, China, in 2009, the M.Sc. degree in information and communication engineering from the Changchun University of Science and Technology, Changchun, in 2013, and the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology, Harbin, China, in 2018.

He is a Post-Doctoral Researcher with the Beijing Remote Sensing Information Institute, Beijing, China. His research interests include artificial intelligence, computer vision, and remote sensing image analysis.



Lihong Kang received the Ph.D. degree in electronic information engineering from the Harbin Institute of Technology, Harbin, China, in 2017.

He is a Professor with the Beijing Remote Sensing Information Institute, Beijing, China. His research interests include electromagnetic scattering computation, calibration of SAR satellites, and remote sensing image analysis.



Xiangwei Xing received the B.Sc. degree in electronic information engineering from the University of Science and Technology of China (USTC), Hefei, China, in 2007, and the M.Sc. degree in photogrammetric and remote sensing and the Ph.D. degree in electronic science and technology from the National University of Defense Technology (NUDT), Changsha, China, in 2009 and 2014, respectively.

His research interests include marine surveillance and pattern recognition.



Jing Tian received the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2007.

She is a Professor with the Beijing Remote Sensing Information Institute, Beijing, China. Her research interests include remote sensing image analysis and artificial intelligence application.



Chunzhuo Fan received the B.Sc. degree in electronic science and technology from the University of Technology of China (USTC), Hefei, China, in 2018.

His research interests include marine surveillance, deep learning, and remote sensing image analysis.



Ye Zhang received the B.S. degree in communication engineering and the M.S. and Ph.D. degrees in communication and electronic system from the Harbin Institute of Technology (HIT), Harbin, China, in 1982, 1985, and 1996, respectively.

From 1998 to 1999, he was a Visiting Scholar with The University of Texas at San Antonio, San Antonio, TX, USA. Since 1985, he has been a Teacher with HIT. He is a Professor and a Doctoral Supervisor in information and communication engineering and the Director of the Institute of Image Engineering, HIT. His research interests are remote sensing hyperspectral image analysis and processing, image/video compression and transmission, as well as multisource information collaboration processing and applications.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:34:26 UTC from IEEE Xplore. Restrictions apply.